Testing Transcripts

Mock Participant 1 -

Researcher: Okay, yeah, so my project is about, like, taking chemistry, lab into VR, and you'll be able to interact with the chemicals and reactants and form a, I mean, start a reaction, that kind of thing. Basically chemistry avenue yard. But like for starters, I want you to, I mean, interact with the reactors and the chemicals that are here and see how it works here.
Okay. Oh, the controls are....

Participant 1: Sorry. When you said, I'm like converting chemistry lab into VR. I want to blow up the lab.

Researcher: Actually, my... No, no, no. Actually, my initial plan started with that because there are a lot of chemicals that we cannot interact in real life, but, you know, we are, we can use those chemicals and see how those work, because as students, we don't get a lot of chemicals to interact with.

Participant 1: Um, it's just, like, how you said that. There would be a technical issue if we don't know how a chemical reacts in real life. How will we simulate it?

Researcher: Uh, I mean, like, um...No, that's actually a good point.
But my plan was, like, a lot of times we study a lot of reactions and chemicals in theory, but we don't know how they work. So, basically, we would know theoretically how it works, but to be sure, like, educational tools. Yeah, yeah, yeah. It will be only for the stuff, which is already, like, tested, and we have... This is how it can be. Yeah, yeah, yeah, yeah.
Thank you. Oh, I'm sorry. Yeah, so these are WASD, and with these, you can use the controller. This is for left controller, right controller, and G for grab.

Okay. So, can you remember any parts that you like?

Participant 1: Oh, sorry. Wait, I I wanna undo it. How do it?
I mean, I even do as a? Like, until, like, this. Oh, yeah.

Researcher: So for that, you'll use this one to select this controller and this one to select this.

Participant 1: But I don't want to select any controller. I just want to move closer.

Researcher: Maybe then... Maybe then you can, uh, wait. I am restarting.
I have close it for... With that time, too. No, but you do…

Participant 1: No, no, with the controller, because... that's supposed to be the control. And grab it, and then what? Actually, there is one error with it.

Researcher: Yeah, yeah, I know. This controller will only grab that reaction, so that I have to rectify. So please ignore that one.
No, I need... This is the beaker where... Okay, I'm just gonna explain really quickly. I'm gonna just... Yeah. Like, a free task.
Yeah, yeah. So the reaction will happen... And right now, it's... Construct ascendants. And make the...
Okay, okay, okay, it's okay. I think that was, like, a problem with the controller. I want you to do so, basically, you just speak your science interaction, right?
And there was a...
I know there's, like, you can, basically, know that molecular focus by expanding it, and, uh, view the molecules, and, uh, it are reduced.

Participant 1: Okay. Okay. Yeah, no, I use my back with the... Maybe so much.

Researcher: Yeah, no, once the controllers are together, then you, like, do the expand feature where, like, they're sitting, and they have modules.
Yeah, and then so that's how you entered over. It has to, it still comes. This one is... It's supposed to be a stable one. You want to reset your view? Oh, you've been instructed on the one.

Participant 1: Okay, but that usually with, like, yeah... Yeah, that... It's not grabbing. Uh, yeah, if you want to say what you're thinking of doing, because...
Okay, so what I understand from this is that I have to pick up the molecules and this is basically supposed to win the right spaces. Yes.

And then the red thing is to like, I'm done. Oh, no, no. Okay. To grab, you need G. Good, G. Oh, yeah. I need to point at it also, right? Yeah.
Yeah. And then how do I control the point?

Researcher: I mean, this is very sensitive right now, but I will be in order, I think, including more, more than you can interact with. But basically, the concept is supposed to be, like, to understand fusion, and, you know, how particles combine together, separated, that kind of thing. And you can also, like, another thing is you can grab this molecule and stretch it. Yeah.

Participant 1: That is very interesting. I know it's... I feel... Since what if a person doesn't break it and hold a molecule? Because I feel like the distance between two bonds would change the chemical compound. Yeah, yeah.
Yeah. Yeah, yeah, yeah. So.

Researcher: So that is how it will work. Like, you can interact with the molecules and see how things change. Because it's supposed to be for educational purposes.

Participant 1: So, yeah. Yeah, yeah. I like that.
How did you find your time? I mean... Okay, um, snapping.
It's just, I would say, the learning curve of using the system. control, control.
Because, um, okay, it's different for everyone. designing it based on themselves.
So for yours, it's like only left control with control, the left one.
The right will control the right points. But, well, no, that's, yeah, that's, that's the only thing I've ever, but I think it's pretty good.

Researcher: Okay, any other ones?

Participant 1: Maybe, okay, I don't know what kind of molecule it is called, or something. Okay, okay. Like more information.
Yeah, about the molecules. Yeah, or maybe in some sense, I am putting the molecule together, maybe I could have a visual, like, representation, how the final molecule should look like. So maybe that...
Okay. Yeah. Oh, yeah. You can throw it away. Oh, yeah, that one's in the right place. Oh.

Mock Participant 2 -

Researcher: So this is my Immersive Chemistry Lab prototype. The idea is that instead of only seeing chemistry on a normal screen, you can like…interact with the experiment and then explore what is happening at the molecular level. I'll give you the basic controller controls, but otherwise just try whatever feels natural…

Participant 2: Okay, sure.

Researcher: You can move around using the controls here, and G is used for grabbing. You have these two reactants and the beaker. So yeah like can you try to start the experiment.

Participant 2: Okay, so this is interesting like I assume these need to go inside the beaker? Maybe umm how do I like…oh okay okay

Researcher: Yeah like yeah Just try what you think would work

Participant 2: Okay. I'm trying to grab this one. Wait, which controller am I moving?

Researcher: This one is currently controlling the left controller.

Participant 2: Oh okay. So I need to get closer first?

Researcher: Yeah, sorry the controller needs to be close enough to the object. I'm sorry I know there's some issues with the controller right now

Participant 2: Okay, got it. Now G?

Researcher: Yep.

Participant 2: Okay, I've got it. Umm okay I'll move it here. And I assume the other one goes there too?

Researcher: Try it, yeah yeah yeah

Participant 2: Okay. Oh, wow something happened.

Researcher: yeah…what do you think happened?

Participant 2: Oh maybe like…I think they reacted because that green thing appeared.

Researcher: Now like this imagine you want to investigate what is happening inside that reaction. What would you try doing here?

Participant 2: I'm trying to touch it with the controller. Nothing is happening. Do I grab it?
Maybe move around it? Oh, like this?

Researcher: Yeah.

Participant 2: Oh what….

Researcher: What do you think this environment represents?

Participant 2: I assume this is maybe umm yeah the reaction, like the atoms or molecules.

Researcher: Yes yes you're right exactly. Now try interacting with what you see.

Participant 2: Okay. I'm assuming I can grab the white ones. Okay, got one. Do I put it near the red one?

Researcher: Do whatever is okay for you

Participant 2: Oh, it snapped in. That's actually pretty clear. Umm so is it like…I formed a bond, I think.

Researcher: Yeah, now try manipulating the molecule.

Participant 2: Can I grab the white one again?

Researcher: Try it.

Participant 2: Oh, the bond is stretching. Okay, if I keep going... Oh, it disappeared.

Researcher: I will just like…okay no this, yeah. So, I just want to know why do you think it disappeared?

Participant 2: Because I pulled the atom too far away, so I broke the bond.

Researcher: Yeah yeah perfect. What was the least obvious interaction?

Participant 2: Definitely getting into this molecular view. I thought the green thing was a button or something I had to select. I felt sorry okay maybe this like stuck?

Researcher: Yeah I can understand it can be confusing….Anything you'd change?

Participant 2: Maybe when the green thing appears, have arrows showing that you need to expand it or move your hands apart. The molecule part made more sense once I got here.

Researcher: Oh yes, I think that's a very good point.
Thank you so much for your time, this was all

Mock Participant 3 -

Researcher: This is an early prototype for an immersive chemistry lab. You're starting at the laboratory scale, and I'd like you to try starting the experiment. I'll explain the simulator controls, but I'll try not to explain the actual interactions. And ah like I made this so that students can experiment with chemicals in a virtual reality and understand how reactions work

Participant 3: Okay….

Researcher: G is grab, and these controls let you manipulate the controllers.

Participant 3: Right. So these like coloured things are chemicals, no?

Researcher: Yeah, they're simplified reactants for this prototype. As this is the first prototype I have kept the UI very very minimal

Participant 3: And this in the middle is the beaker where I mix the other two?

Researcher: Yes

Participant 3: Okay, I'll grab this one. Umm oh no how do I pick this? Oh wait, it's not grabbing.

Researcher: yeah maybe try moving the controller closer first

Participant 3: Oh okay. Now it grabbed now

Researcher: Yep.

Participant 3: I'll put it here. Now the other one. Umm I feel yeah..I feel this is a bit awkward that I need to use different controllers for different chemicals, you know like, yeah okay

Researcher: Oh true, okay I understood, Controllers do need some fixing I understand

Participant 3: Oh. Wait, what happened? I moved this like this my controller back and everything changed.

Researcher: Yeah so like yes this is expected to happen. Perfect so, umm what did you think caused that?

Participant 3: okay….I honestly don't know. I thought I was just moving away from the beaker and suddenly things changed a lot like this

Researcher: What do you think this new environment is?

Participant 3: Because there are atoms so maybe like something to do with molecules and inside the reaction or something I think…

Researcher: Yes, actually you're right about this, good good

Participant 3: Okay. I understood where I ended up, but I didn't understand what I did to get here.

Researcher: Yeah right I can understand that, that's useful. Now umm can you like try interacting with the molecule.

Participant 3: I'm trying to grab one of these white ones…yeah this okay

Researcher: Okay.

Participant 3: It's a little so this is difficult getting the controller exactly where I want it.

Researcher: That's okay, you can keep on trying maybe do this…like now press G and then here…

Participant 3: Got it…so if I do this will it..Oh, it just connected.

Researcher: Yes perfect what do you think happened?

Participant 3: It automatically puts it into the correct place near red one, which is something I like because this yes getting it exact with these controls would be difficult.

Researcher: True, I agree. Try manipulating it now.

Participant 3: Maybe rotate it? Hmm okay I can't pick this one okay I'll grab the white atom again.

Researcher: Okay.

Participant 3: Oh, that's interesting. The connector gets longer and stretches out wow

Researcher: What would you try next?

Participant 3: I dont know maybe this like keep pulling it?

Researcher: sure

Participant 3: And it broke. Okay. That makes sense umm yeah okay

Researcher: okay now I will just ask you few questions like umm are you okay with this? Which interaction was most natural like you didn't need my explanations?

Participant 3: Putting the atom near the other atom and then pulling it…like you know..like stretch it like this. That was pretty obvious.

Researcher: yeah yeah nice and umm least natural?

Participant 3: The transition to this one this place. It happened without me intentionally doing anything with these. Umm maybe you can you know probably be something that confirms I actually want to enter that mode.

Researcher: Yeah I will try to fix this part, right right. Oh okay Any other feedback?

Participant 3: Umm not much like maybe some text when the reaction happens, like "Explore molecular view" or something. Otherwise you don't really know what the green object means and it's like there and it's floating and appears like this, if you know what i mean , right

Researcher: Yes yeah that's very true. Okay thank you for the interview. I will end it here, okay?

Mock Participant 4

Researcher: Okay so my project is an XR chemistry lab where you perform a simplified experiment like you know sometimes students like umm can't perform some experiments in real life like chemicals are not available maybe so this project lets students perform experiments in this like virtual reality and then move into molecular scale to interact directly with the molecule. Here, I'm testing the interactions rather than your chemistry knowledge.

Participant 4: Oh okay, thats cool

Researcher: These are the simulator controls, if you have used VR headset before then you know these are the controllers for it. You can move the controllers with here these keys like this and and use G to grab objects.

Participant 4: So can I use either controller with this right here umm

Researcher: Yes that's okay now umm try interacting with the objects and tell me what you expect.

Participant 4: Okay. I'll use this one here okay what is this? This is so awkward to use oh
It's not grabbing like why

Researcher: oh yes maybe umm try the other controller.

Participant 4: Oh, okay that one works. Why does only that one work here? I think this is uncomfortable.

Researcher: Yeah oh umm I agree. That's actually a limitation in this version. Some of the objects are currently assigned to a particular controller here like

Participant 4: Oh. That's confusing because in VR I'd expect either hand to work na?

Researcher: Yeah true I will improve that actually. Can you keep going with the working controller for now maybe I'm sorry

Participant 4: Okay. I'll move this toward the beaker.

Researcher: Yep, ohh

Participant 4: And I need the other reactant too?

Researcher: Yeah like just try what you think.

Participant 4: Okay. There is now umm there's a green ball or something on it. What is this

Researcher: What do you think you should do next?

Participant 4: Maybe grab it…where do you grab oh umm yeah G

Researcher: Yeah sorry about the controls but for now like try whatever seems natural.

Participant 4: This doesn't seem to grab. Maybe if I just can I move both hands…this is how do I move both hands. Okay holding this maybe

Researcher: Okay so I will help you a little here try moving the controllers around the focus and then like this you know outward.

Participant 4: Like zooming?

Researcher: yeah yeah good try it.

Participant 4: Oh, okay. Now I'm inside it. That's actually cool. So it's like zooming into the reaction?

Researcher: right that's correct cause that's the intention.

Participant 4: Okay, that makes more sense now…umm okay now what can I do here

Researcher: Try completing the molecule or umm playing with it like play around with it yeah yeah

Participant 4: So I grab the white atom is this okay

Researcher: Yep, try it.

Participant 4: Again, which controller?

Researcher: It's okay so try the one you were using before.

Participant 4: Yeah, that's probably my main issue. I don't want to have to remember which hand works with which thing cause umm this is so awkwards like if you only use certain hands, getting it

Researcher: Okay yeah yeah I agree

Participant 4: Now ohh oh..it's snapped in. That part is really good.

Researcher: Yeah right okay umm try doing something else with the molecule. Like if you're in a chemistry lab experiment umm how like how will you play with it yeah yeah

Participant 4: oh so I'll pull this atom back out. Oh, the bond stretches. And... okay,
Now it can break okay

Researcher: Yeah okay this was actually it so I will ask you some questions now. What would you change first umm if this was your project.

Participant 4: Definitely make both hands work for everything. In VR I wouldn't think like this object belongs to my left hand. I'd just reach for it with whichever hand is closer like here you know

Researcher: Yeah true so umm anything else you'd like to include?

Participant 4: Maybe make the lab look more like a lab eventually, but interaction-wise the controller thing is the biggest issue for me.

Mock Participant 5 -

Researcher: This is my Immersive Chemistry Lab prototype. You start by interacting with reactants at the laboratory level, and the idea is that you can then explore the same reaction at molecular scale. I'm mainly testing whether the interaction sequence makes sense.

Participant 5: Okay, sounds good.

Researcher: These are the simulator controls, and G is grab. Try starting the experiment.

Participant 5: So I have two reactants and a beaker. I assume I'm combining these.

Researcher: Try it.

Participant 5: Okay, let me grab this one. The controller is a little awkward.

Researcher: Yeah, it is being tested through the simulator at the moment.

Participant 5: Okay. First one is in. I'll get the second one.

Participant 5: There. Oh, green sphere.

Researcher: What do you think that means?

Participant 5: Maybe it's showing the reaction happened? Or maybe I can interact with it?

Researcher: Imagine you want to investigate what's happening inside the reaction.

Participant 5: Maybe expand it? Like zoom in?

Researcher: Try that.

Participant 5: Oh, there we go.

Researcher: What do you think happened?

Participant 5: I've gone from the normal lab into a microscopic or molecular view of the reaction.

Researcher: Now try interacting with the atoms.

Participant 5: Okay. Red is probably Oxygen and white is Hydrogen?

Researcher: Yes.

Participant 5: Am I supposed to make water?

Researcher: Try completing the molecule.

Participant 5: Okay, I'll move this Hydrogen here.

Participant 5: Oh, it snaps. That's useful.

Researcher: What did that communicate to you?

Participant 5: That I've put the atom into a valid position and formed the bond.

Researcher: And what else would you try?

Participant 5: I'll grab it again. Oh, the bond stretches.

Researcher: Keep trying what feels natural.

Participant 5: And it breaks when I go too far. Yeah, I understand that.

Researcher: Which interaction did you find most natural?

Participant 5: Probably grabbing the atoms and putting them together.

Researcher: What was confusing?

Participant 5: I understood it once I saw the atoms, but I think you need more information about what molecule I'm actually building.

Researcher: What kind of information?

Participant 5: Maybe show the name of the molecule and maybe a small diagram of what the completed structure should look like. Then I'm not just randomly connecting things.

Researcher: Okay.

Participant 5: Especially if it's educational. It could tell me what bond I'm making or why it broke when I moved it too far.

Researcher: Anything about the transition?

Participant 5: I actually liked the idea of zooming into the reaction. Maybe just make the cue for expanding more obvious.

Researcher: Anything else you'd change?

Participant 5: Mostly instructions and information. The interaction makes sense once you know what you're supposed to build.